

Roboguide-ROS2-RViz Integration

FANUC RoboGuide ↔ ROS 2 ↔ RViz Digital Twin Integration Guide

Environment: Windows 11 + WSL2 (Ubuntu 22.04) + FANUC RoboGuide (HandlingPRO)

ROS 2 Distribution: Humble Hawksbill

Robot Model: CRX-10iA/L

Driver: FANUC ROS 2 Driver v1.1.5 (installed via Debian packages)

Prepared by: Himanshu Sharma

Version: 1.0

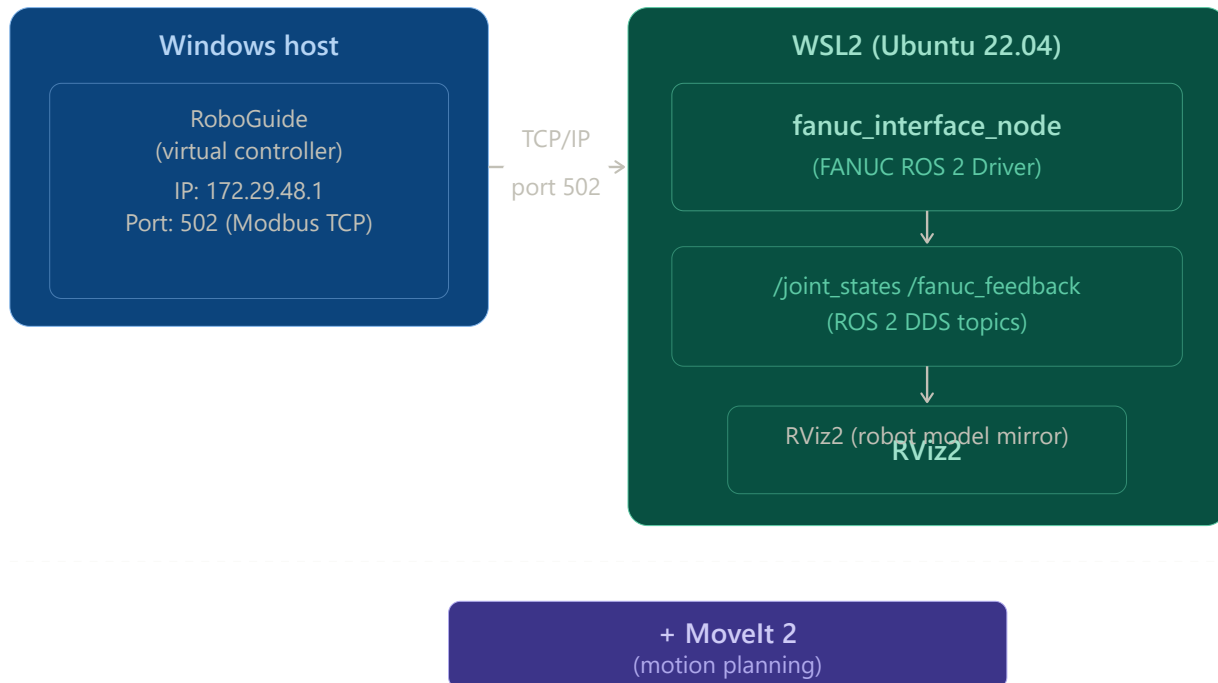
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1. Architecture Overview

The integration creates a **bidirectional digital twin**: jogging the robot in RoboGuide moves the RViz model in real time, and planning/executing trajectories in RViz drives the RoboGuide virtual controller.

Architecture overview



Data flow

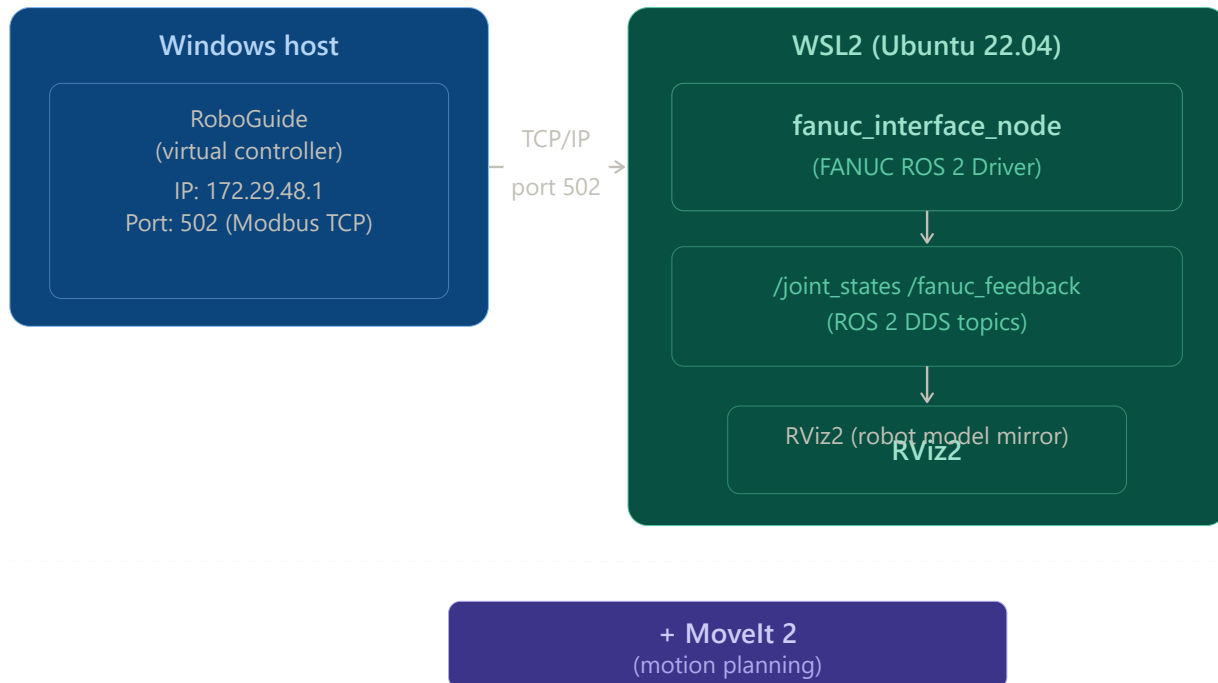
- RoboGuide → TCP/IP port 502 → FANUC ROS 2 Driver → /joint_states → RViz (mirror)
- RViz interactive marker → MoveIt 2 plan → /follow_joint_trajectory → FANUC driver → TCP/IP → RoboGuide (execute)

The IP address for you could be different

2. Prerequisites

Component	Requirement
OS (Windows)	Windows 10/11 with WSL2 enabled
OS (Linux)	Ubuntu 22.04 inside WSL2
ROS 2	Humble Hawksbill
FANUC Software	RoboGuide (HandlingPRO), R-30iB Plus controller, software version V9.40 or higher
Robot Option	ROS 2 Package (S568) must be selected during workcell creation
FANUC Driver	fanuc_ros2_driver_v1_1_5.zip (Download it here)
Network	WSL2 must be able to ping the RoboGuide virtual controller IP

Architecture overview



Data flow

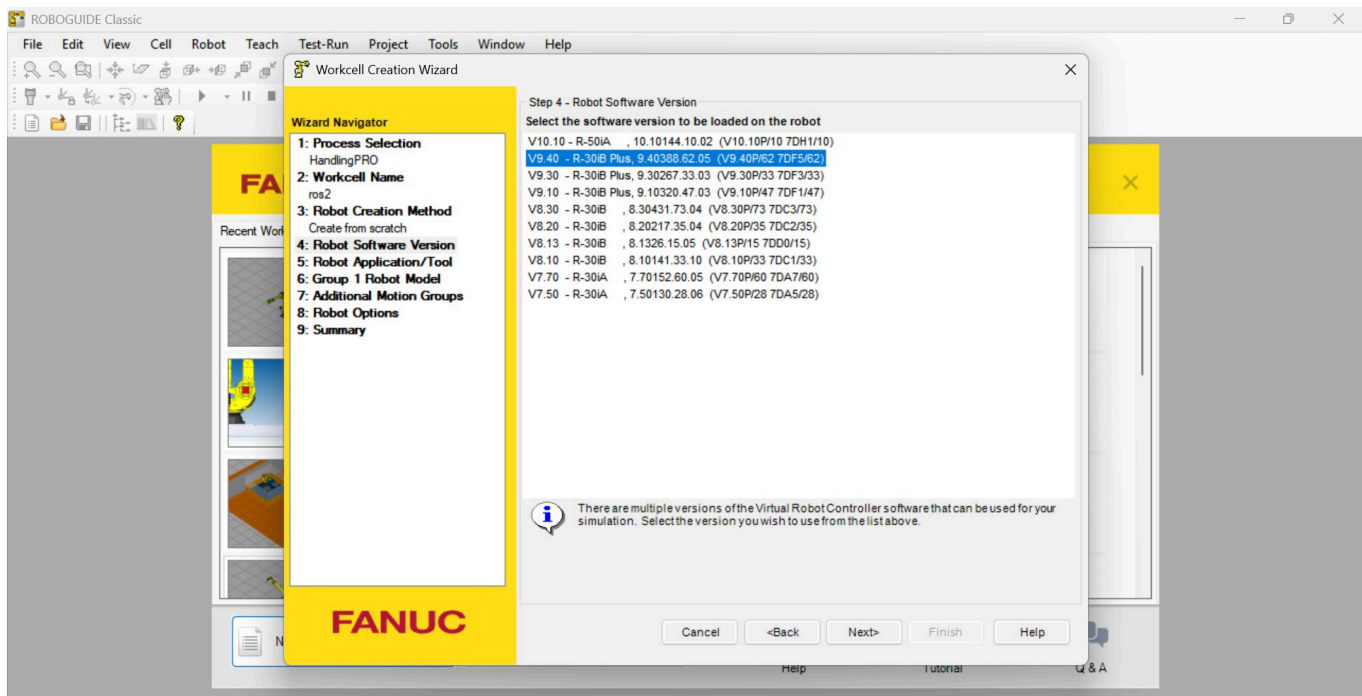
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3. Part A — RoboGuide Setup (Windows)

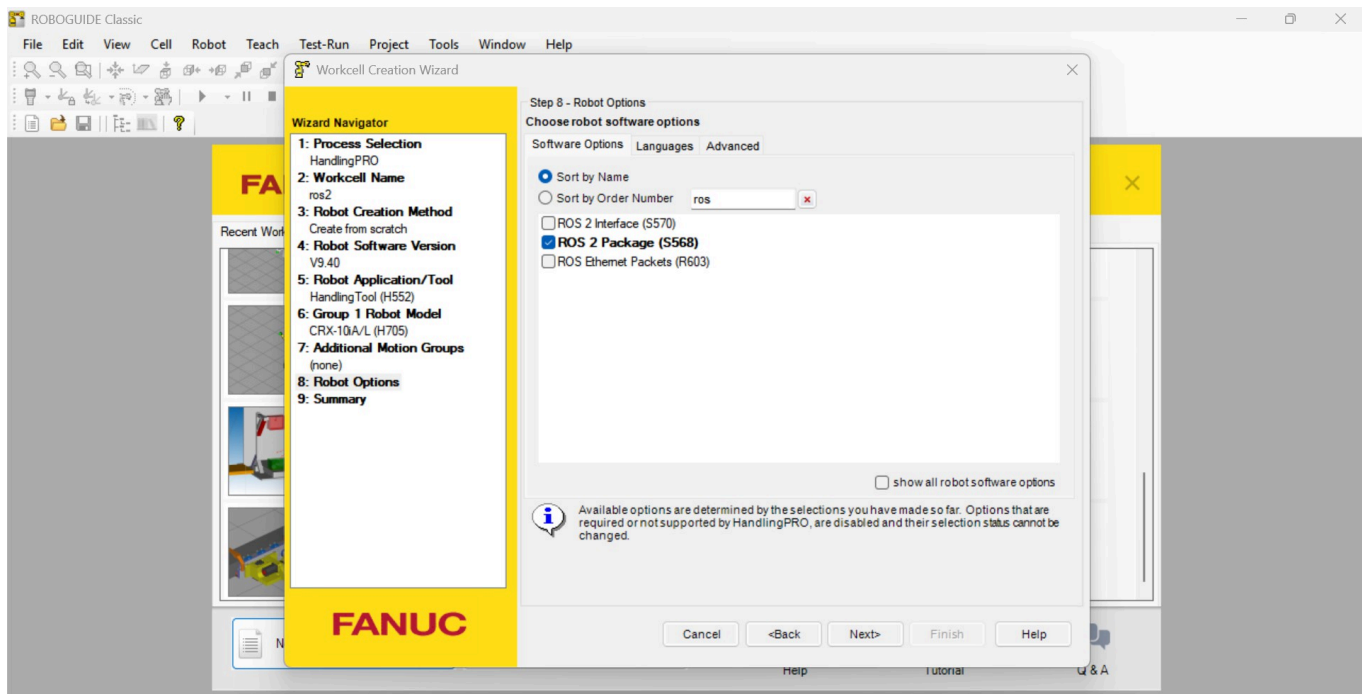
3.1 Create the Workcell with Correct Options

When creating a new HandlingPRO workcell in RoboGuide:

Step 1 – Robot Software Version: Select **V9.40** (R-30iB Plus) or higher — this is required for ROS 2 driver compatibility.



Step 2 – Robot Options: In the Software Options tab, search for `ros` and enable:
- **ROS 2 Package (\$568)** ← this is the critical one

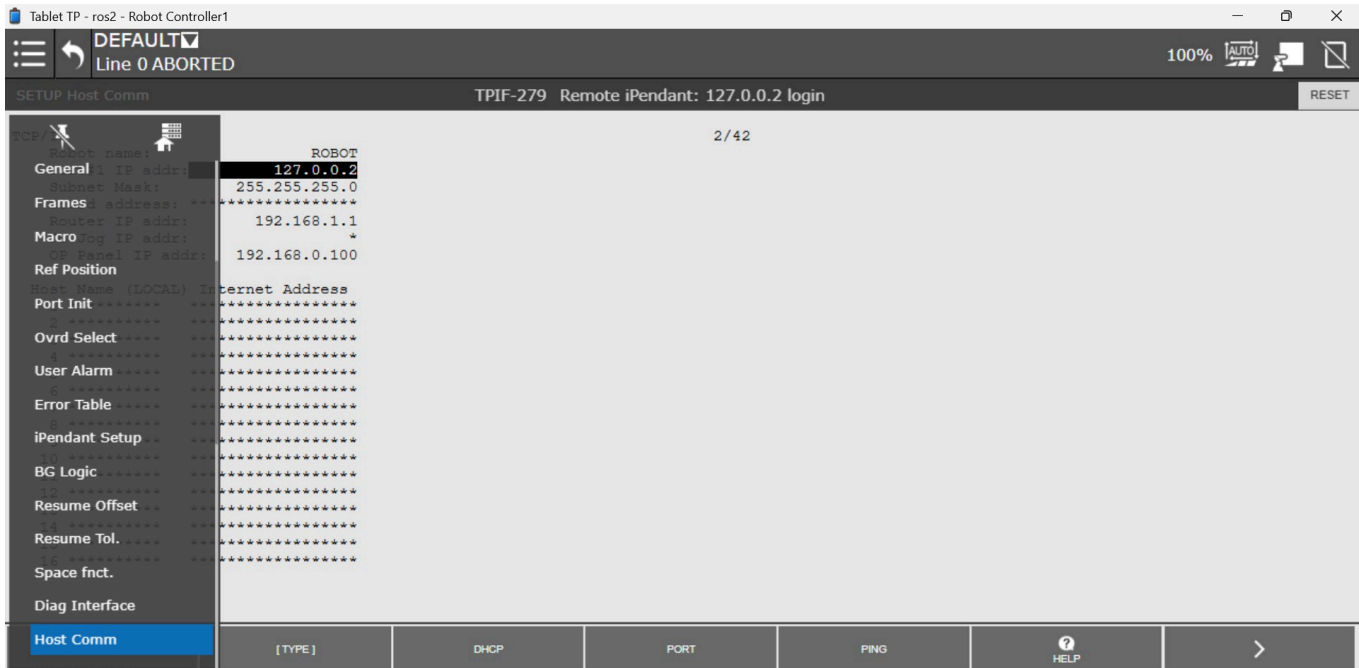


Step 3 – Complete the wizard and start the work cell.

3.2 Set the Controller IP Address

The RoboGuide virtual controller needs an IP address that WSL2 can reach. By default it may be set to `127.0.0.1` (loopback) — this will **not** work from WSL2.

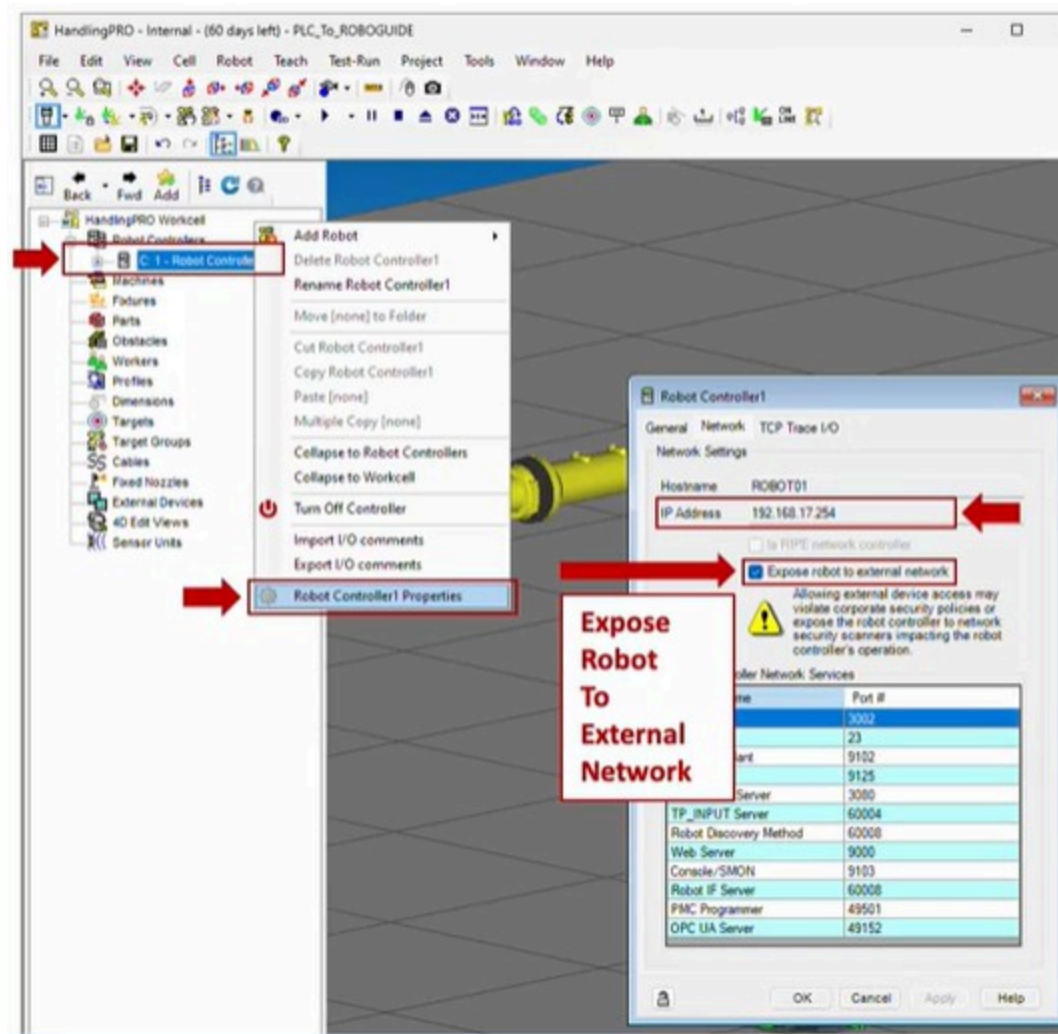
1. In RoboGuide, open the virtual teach pendant: **Robot** → **Teach Pendant** (or **Robot** → **Remote iPendant**)
2. On the teach pendant, navigate to: **MENU** → **Setup** → **Host Comm**



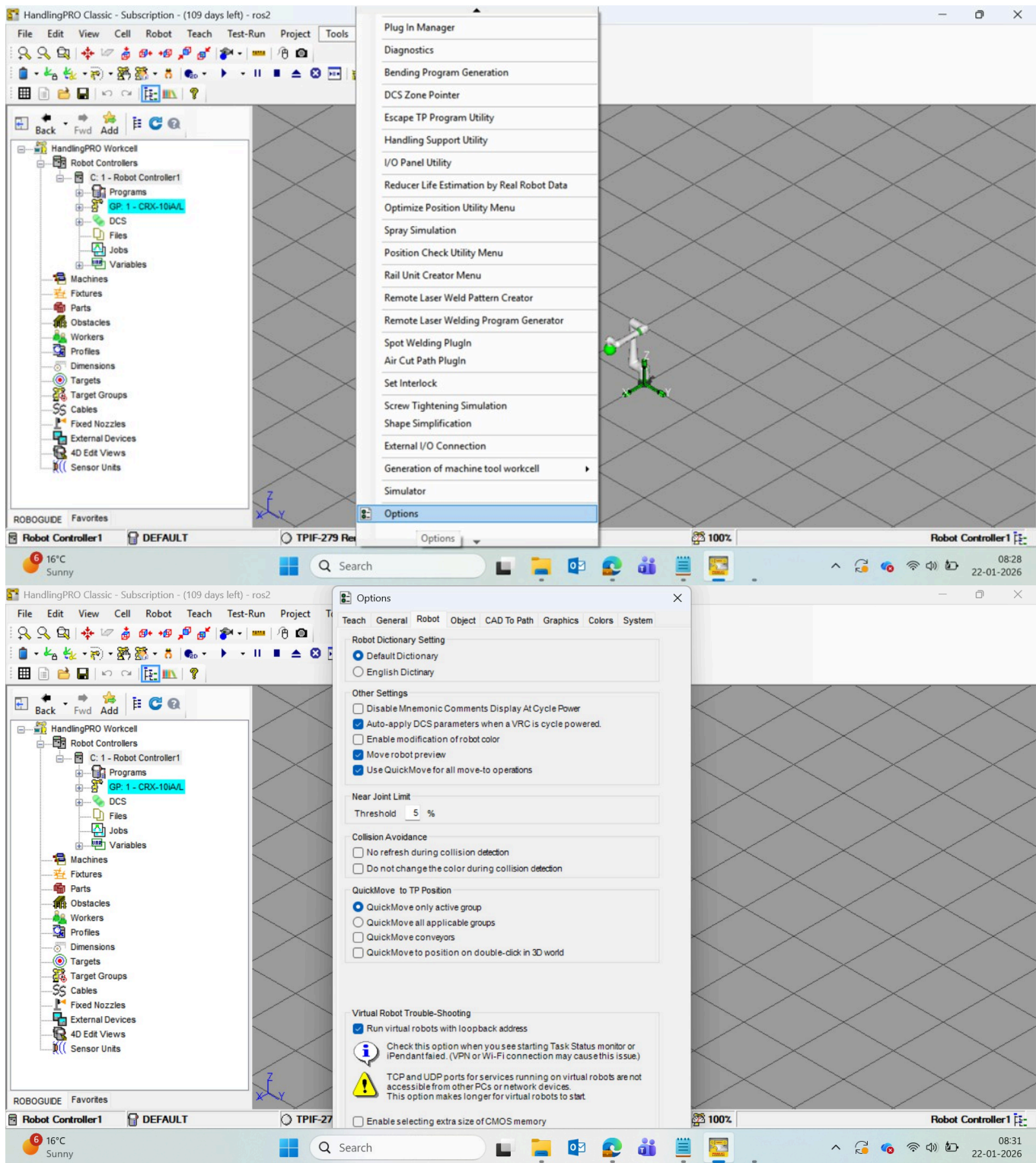
3. Find **Port#1 IP addr** — change it from 127.0.0.1 to your **Windows host IP** as seen from WSL2 (e.g. 172.29.48.1). See [Section 6](#) for how to find this IP.
4. Set **Subnet Mask** to 255.255.0.0 (for a 172.x.x network — **not** 255.255.255.0 or routing will fail)
5. Press **FCTN** → **Cold Start** to apply the IP change (a cold start is required)

3.3 Expose the Controller to External Network

Right-click **Robot Controller1** in the workcell tree → **Robot Controller1 Properties** → **Network** tab → check "**Expose robot to external network**" → OK.



If "Expose robot to external network" is greyed out or fails: Go to **Tools** → **Options** → **Robot** tab → under *Virtual Robot Trouble-Shooting*, enable **"Run virtual robots with loopback address"** — then retry.



3.4 Verify Modbus TCP is Enabled

After the cold start, on the teach pendant:

1. Navigate to **MENU** → **Setup** → **Host Comm**
2. Press the **>** button at the bottom to scroll right to the **Modbus TCP** page
3. Confirm it shows **Enabled** and is set to **port 502**

4. Part B — WSL2 Environment Setup

4.1 Update System and Install Basics

```
sudo apt update && sudo apt -y upgrade
sudo apt -y install curl gnupg2 lsb-release build-essential git unzip
```

4.2 Install ROS 2 Humble

```
# Add ROS 2 apt repository
sudo apt -y install software-properties-common
sudo add-apt-repository universe
sudo apt -y install curl
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key \
  -o /etc/apt/keyrings/ros-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/ros-archive-
keyring.gpg] \
  http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main"
\
  | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
sudo apt update

# Install desktop + required packages
sudo apt -y install ros-humble-desktop ros-humble-rviz2 \
  ros-humble-control-msgs ros-humble-sensor-msgs

# Auto-source ROS on every new terminal
echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

4.3 Install System Dependencies for the FANUC Driver

```
sudo apt -y install libmodbus-dev libjsoncpp-dev
```

5. Part C — Install FANUC ROS 2 Driver

The driver zip (`fanuc_ros2_driver_v1_1_5.zip`) contains several folders. Since you're on Ubuntu 22.04 (Jammy) + Humble, use the **Debian package method** — it's simpler and doesn't require building from source.

5.1 Copy the Zip into WSL2

From your WSL2 terminal (assuming the zip is in Windows Downloads):


```
cp /mnt/c/Users/<your_windows_username>/Downloads/fanuc_ros2_driver_v1_1_5.zip ~/
cd ~
sudo apt install unzip -y
unzip fanuc_ros2_driver_v1_1_5.zip
```

5.2 Install via Debian Packages

```
cd ~/fanuc_ros2_driver_v1_1_5
unzip install_humble_jammy.zip
cd debs_humble_jammy      # or debs_humble depending on zip structure
sudo dpkg -i *
```

The Debian package method installs driver packages directly to the system path. No need to source a `local_setup.bash` — `source /opt/ros/humble/setup.bash` is all you need.

5.3 Verify Installation

```
source /opt/ros/humble/setup.bash
ros2 pkg list | grep fanuc
```

You should see packages like `fanuc_ros2_driver`, `fanuc_crx_description`, `fanuc_moveit_config`, etc.

6. Part D — Network Configuration (WSL2 ↔ RoboGuide)

This is the most common failure point. WSL2 uses NAT by default, so RoboGuide on Windows and the ROS 2 driver in WSL2 need a port forwarding rule to communicate.

6.1 Find Your IPs

In your **WSL2 terminal**:

```
# Your WSL2 IP
ip addr show eth0 | grep "inet "

# Your Windows host IP (the gateway WSL2 uses - this is what RoboGuide binds to)
ip route | grep default
# e.g. "default via 172.29.48.1 dev eth0" → Windows host IP = 172.29.48.1
```

In **Windows CMD or PowerShell**:

```
ipconfig
# Look for "WSL" adapter - typically 172.x.x.1
```

6.2 Test Connectivity

From WSL2, test if port 502 on the Windows host is reachable:

```
nc -zv 172.29.48.1 502
```

- **"Connection refused"** → RoboGuide Modbus TCP is not running or the controller IP is wrong. Go back to Section 3.2–3.4.
- **"Blinks / times out"** → Windows Firewall is blocking port 502. Continue below.
- **"succeeded"** → You're good, skip to Section 7.

6.3 Fix Windows Firewall (run once as Administrator)

Open **PowerShell as Administrator** on Windows:

```
# Check if anything is listening on port 502 (should show RoboGuide)
netstat -ano | findstr :502

# Add inbound firewall rule for Modbus TCP
netsh advfirewall firewall add rule name="Modbus TCP 502" dir=in action=allow
protocol=TCP localport=502

# Add outbound rule too
netsh advfirewall firewall add rule name="Modbus502out" dir=out action=allow protocol=TCP
localport=502
```

Re-test from WSL2:

```
nc -zv 172.29.48.1 502
# Should now say: Connection to 172.29.48.1 502 port [tcp/*] succeeded!
```

If it still times out after the firewall rule, temporarily disable the firewall completely to confirm it's the culprit:

```
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False
```

Test again. If it now connects, the firewall was the issue — re-enable it after confirming the rule is in place:

```
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled True
```

6.4 Update RoboGuide Controller IP (if needed)

If the controller IP is still 127.0.0.1 from defaults, it will never be reachable from WSL2. As described in Section 3.2, change it on the teach pendant to 172.29.48.1 (your Windows host IP as seen from WSL2), then Cold Start.

7. Part E — Launch the Driver and Verify Connection

7.1 Launch the FANUC Interface Node

In a WSL2 terminal (Terminal 1):

```
source /opt/ros/humble/setup.bash

ros2 launch fanuc_ros2_driver fanuc_interface.launch.py \
  robot_ip:="172.29.48.1" \
  robot_type:="crx10ia_l"
```

Replace `172.29.48.1` with your actual Windows host IP from Section 6.1.

Success looks like this in the terminal output:

```
[fanuc_interface_node-1] [INFO] [fanuc_interface]: [fanuc] robot controller's ip address :
172.29.48.1
[fanuc_interface_node-1] [INFO] [fanuc_interface]: [fanuc] robot controller's port
: 502
[fanuc_interface_node-1] [INFO] [fanuc_interface]: [fanuc] robot type
: crx10ia_l
[fanuc_interface_node-1] [INFO] [fanuc_interface]: [fanuc] RMI version: Major 7 Minor 0
[fanuc_interface_node-1] [INFO] [fanuc_interface]: [fanuc] Communication is established :
16002
```

The key line is `Communication is established : 16002` — that's your green light.

If port 502 is already in use (e.g. by R800 Modbus TCP): add `robot_port:=503` to the launch command.

7.2 Verify ROS 2 Topics are Publishing

In a second WSL2 terminal (Terminal 2):

```
source /opt/ros/humble/setup.bash

ros2 topic list
# Should show at minimum:
# /joint_states
# /fanuc_feedback
# /parameter_events
# /rosout

# Watch live joint data
ros2 topic echo /joint_states
```

Jog the robot in RoboGuide (**Robot** → **Show Joint Jog Tool** → drag the J1–J6 sliders). You should see the joint values updating in real time in the terminal.

8. Part F — Launch RViz Digital Twin

8.1 Install MoveIt 2 (one-time)

```
sudo apt install -y \  
  ros-humble-moveit \  
  ros-humble-moveit-ros-visualization \  
  ros-humble-moveit-setup-assistant \  
  ros-humble-joint-state-publisher-gui \  
  ros-humble-robot-state-publisher \  
  ros-humble-xacro \  
  ros-humble-ros2-control \  
  ros-humble-ros2-controllers
```

8.2 Launch RViz with MoveIt 2

Keep Terminal 1 (driver) running. Open Terminal 2:

```
source /opt/ros/humble/setup.bash  
  
ros2 launch fanuc_ros2_driver test_fanuc.launch.py robot_type:="crx10ia_l"
```

This opens RViz with:

- The **CRX-10iA/L robot model** loaded
- The **MotionPlanning panel** on the left
- **Interactive markers** (coloured arrows/rings) at the robot TCP

8.3 Confirm the Mirror is Live

Now jog the robot in RoboGuide (**Robot** → **Show Joint Jog Tool**). The **orange robot arm in RViz should move in real time** to match.

To double-check data flow in Terminal 3:

```
source /opt/ros/humble/setup.bash  
ros2 topic echo /joint_states
```

Joint values should update as you jog.

What you're seeing in RViz:

- **Orange robot** = current live position (from `/joint_states` via RoboGuide)
 - **Grey ghost robot** = planned target position (from the interactive marker)
-

9. Part G — Two-Way Control (Plan & Execute from RViz)

This sends trajectories from RViz back to RoboGuide — the robot physically executes the motion you plan.

9.1 Prepare RoboGuide Controller

The controller must be in **AUTO + Remote mode** to accept trajectory commands from ROS 2. On the virtual teach pendant:

1. **Robot → Teach Pendant** (or **Robot → Remote iPendant**)
2. Find the **3-position key/mode switch** (bottom-left of pendant) — rotate it from **T2** to **AUTO**
3. Press **RESET** to clear any ABORTED state
4. Press **SHIFT + RESET** to turn servo ON
5. Press the **top-right square button** until it shows a **diagonal line** (this enables remote mode)

The status bar should show: **AUTO** (green), servo ON, no ABORTED, Run indicator green.

9.2 Plan and Execute from RViz

With both terminals still running (driver + RViz):

1. In RViz, click **"Interact"** at the top toolbar
2. **Drag the coloured arrows/rings** at the robot TCP to set a goal pose — the grey ghost shows where you're sending it
3. In the **MotionPlanning** left panel → **Planning** tab:
 - Click **"Plan"** — you'll see a preview animation (orange arm moves to grey)
 - Click **"Execute"** (or **"Plan & Execute"** directly)
4. Watch RoboGuide execute the exact same motion

If Execute fails with "Failed" in the MotionPlanning panel: the controller is likely still in T2/ABORTED mode. Repeat Section 9.1.

10. Daily Startup Procedure

Once everything is set up, this is all you need each session:

On Windows:

1. Open RoboGuide and start the workcell
2. Confirm the virtual controller is running (green status)
3. Confirm Modbus TCP is enabled on port 502 (MENU → Setup → Host Comm → scroll right)

In WSL2 — Terminal 1 (driver):

```
source /opt/ros/humble/setup.bash
ros2 launch fanuc_ros2_driver fanuc_interface.launch.py \
```

```
robot_ip:="172.29.48.1" \  
robot_type:="crx10ia_l"  
# Wait for: Communication is established : 16002
```

In WSL2 — Terminal 2 (RViz):

```
source /opt/ros/humble/setup.bash  
ros2 launch fanuc_ros2_driver test_fanuc.launch.py robot_type:="crx10ia_l"
```

Jog robot in RoboGuide → RViz mirrors it. Plan in RViz → RoboGuide executes it.

11. Troubleshooting

Symptom	Likely Cause	Fix
Communication is established never appears	Wrong robot IP or port 502 blocked	Check firewall (Section 6.3), confirm controller IP is not 127.0.0.1 (Section 3.2)
nc -zv <ip> 502 times out / blinks	Windows Firewall blocking	Run the netsh firewall commands in Section 6.3
nc -zv <ip> 502 says "Connection refused"	Modbus TCP not enabled in RoboGuide, or controller not started	Check MENU → Setup → Host Comm → Modbus TCP page; ensure controller is green
ros2: command not found	ROS not sourced	Run source /opt/ros/humble/setup.bash
RViz opens but robot doesn't move when jogging	Driver not connected / wrong topic	Confirm Terminal 1 shows established : 16002 ; check ros2 topic echo /joint_states
Plan & Execute returns "Failed" in RViz	Controller in T2 or ABORTED	Switch to AUTO mode + RESET + SHIFT+RESET on teach pendant (Section 9.1)
source ~/fanuc_ros2_driver/install/local_setup.bash — No such file	Driver installed via Debian packages, not ZIP	Don't source local_setup.bash — just source /opt/ros/humble/setup.bash is enough
fanuc_libs "not found" warning during build	Harmless CMake warning	Ignore it — RViz launches fine anyway

Symptom	Likely Cause	Fix
RoboGuide IP still 127.0.0.1 after cold start	Cold start not completed	Press FCTN → Cold Start and wait fully; check Host Comm again after restart
Firewall rule added but nc still times out	Windows Defender overriding rules	Temporarily set Set-NetFirewallProfile -Enabled False , test, then re-enable

Appendix: Key Commands Reference

```
# Source ROS 2 (run in every new terminal)
source /opt/ros/humble/setup.bash

# Launch FANUC driver
ros2 launch fanuc_ros2_driver fanuc_interface.launch.py \
  robot_ip:="<WINDOWS_HOST_IP>" robot_type:="crx10ia_l"

# Launch RViz + MoveIt digital twin
ros2 launch fanuc_ros2_driver test_fanuc.launch.py robot_type:="crx10ia_l"

# Check topics
ros2 topic list

# Watch live joint data
ros2 topic echo /joint_states

# Find your Windows host IP from WSL2
ip route | grep default

# Test port connectivity
nc -zv <ip> 502

# Firewall rule (Windows PowerShell as Admin)
netsh advfirewall firewall add rule name="Modbus TCP 502" dir=in action=allow
protocol=TCP localport=502
```

This document was assembled from the FANUC India ROS 2 Driver Training Guide (Nikhil sir, v1.0), the official FANUC India setup procedure email (Manoj D Sir., Feb 2026), and the integration chat sessions that produced the working WSL2 setup.